

“We have always gone towards higher and higher integration. Starting from small chips, we kept on integrating more features and blocks into it. From just individual sensors in the beginning, we brought in more sensors into the mix to create a sensor hub for engineers. Embedded micro-controllers and a communication system followed this, and the final result is the self-contained unit we see today,” adds Sharma.

Large-area integration of these micro-electronics devices makes the most of their small size by creating processes that can scale these over a larger area.

ISORG and Plastic Logic have a flexible plastic image sensor made from flexible photodetector sensors and organic thin-film technology. This technology won them the prestigious FLEXI award, a recognition given to firms in the flexible and printed electronics segment.

Integrated smartsensors

Newer sensor technology has now integrated vital components of a smartsensor on a chip to offer a controlled specification set across the operation range of a sensor. The underlying idea here is to integrate sensor technology at the silicon level itself. This is believed to improve power consumption while simplifying product development. How are companies implementing it?

Texas Instruments not only integrates data conversion and communication sections of a smartsensor but also helps in either eliminating the traditional sensing element or integrating it on-chip. For instance, a wheel speed sensor typically employs either a multi-pole ring magnet and hall arrangement or a magnetic rotary encoder and magnet arrangement for measuring wheel speed in vehicles.

What is a sensor

The definition of a smartsensor may vary but, typically, at a minimum, it is the combination of a sensing element with processing capabilities provided by a microcontroller. This essentially means that smartsensors are basic sensing elements with embedded intelligence. The sensor signal is fed to the microcontroller, which processes the data and provides an informative output to an external user. Upcoming trends for smartsensors also include integration of a sensor element and related processing at chip level, thereby making end-product realisation easier and faster to develop.

—Sanjay Jain, analogue applications manager, MGTS, Texas Instruments India (SC sales and marketing)



“The proposed inductive switch sensor completely does away with a costly multi-pole ring magnet and utilises the metallic wheel hub and printed circuit board itself as a sensor to measure wheel speed. Inductive sensing utilises LC tank resonance to identify presence of metallic teeth and valley as an object to switch between high/low states. Given sensitivity, mounting as well as temperature issues with magnets, new solutions make way for a reliable non-magnet approach and low-cost implementation. Moreover, this technology is enabling the placement of control electronics remotely from location of sensing, thereby making it easier to operate the sub-system away from noise environment,” explains Sanjay Jain, analogue applications manager, MGTS, Texas Instruments India (SC sales and marketing).

This technology is also being adapted for a whole lot of other position and speed-sensing applications including passenger-occupancy detection, seatbelt-buckle detection and gear-position detection, to name a few.

Allegro Micro Systems had launched an angle-position sensor that is also programmable. Model A1335 is a contactless, programmable magnetic angle position sensor integrated circuit that comes with a system-on-chip architecture with a front-end based on circular vertical

hall (CVH) technology, programmable microprocessor based signal processing and supports multiple communication interfaces including inter-integrated circuit, serial peripheral interface and single-edge nibble transmission.

Handling data

Traditional heavy cryptography is difficult to deploy on a typical sensor, hence deployment of many insecure IoT devices. “Regulations for the IoT need to address issues of minimum specifications for IoT devices. Existing IoT sensors are not equipped to take advantage of 5G technology either,” explained Kevin Curran, IEEE senior member and reader in computer science at University of Ulster in an interview with EFY.

Production process upgrades

In an interaction earlier this year with Uday Prabhu, general manager, electronics product engineering and product management, Robert Bosch Engineering and Business Solutions Pvt Ltd, he explained that there is compression happening on a large scale with MEMS. An increase in the number of transistors that can go into a chip has resulted in reduced size with higher functionality. Earlier, MEMS elements functioned exclusively as sensing elements.

“Now, intelligence, which is the processor, is built into the MEMS